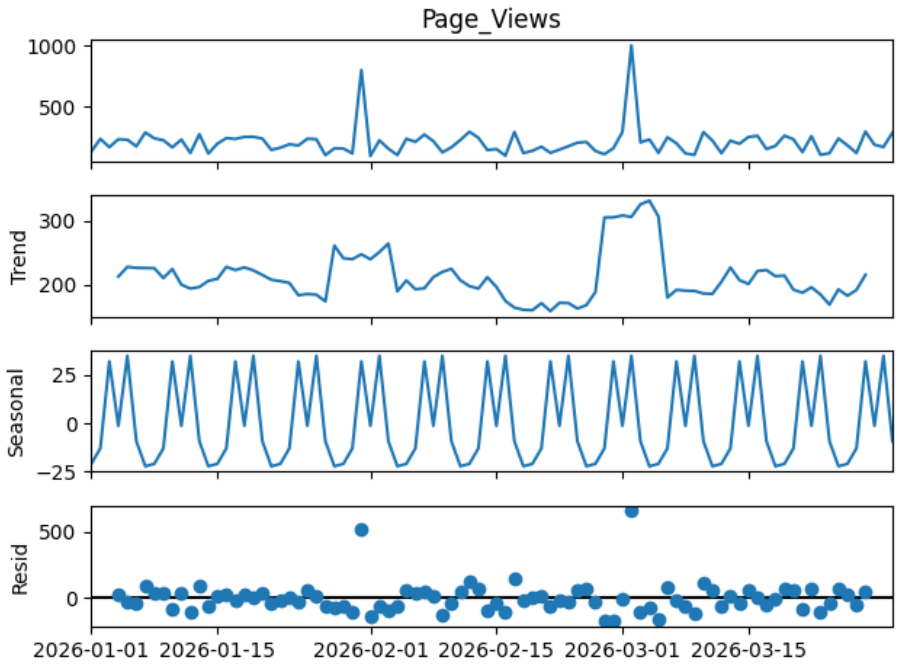
16) Time Series Analysis of Website Traffic.

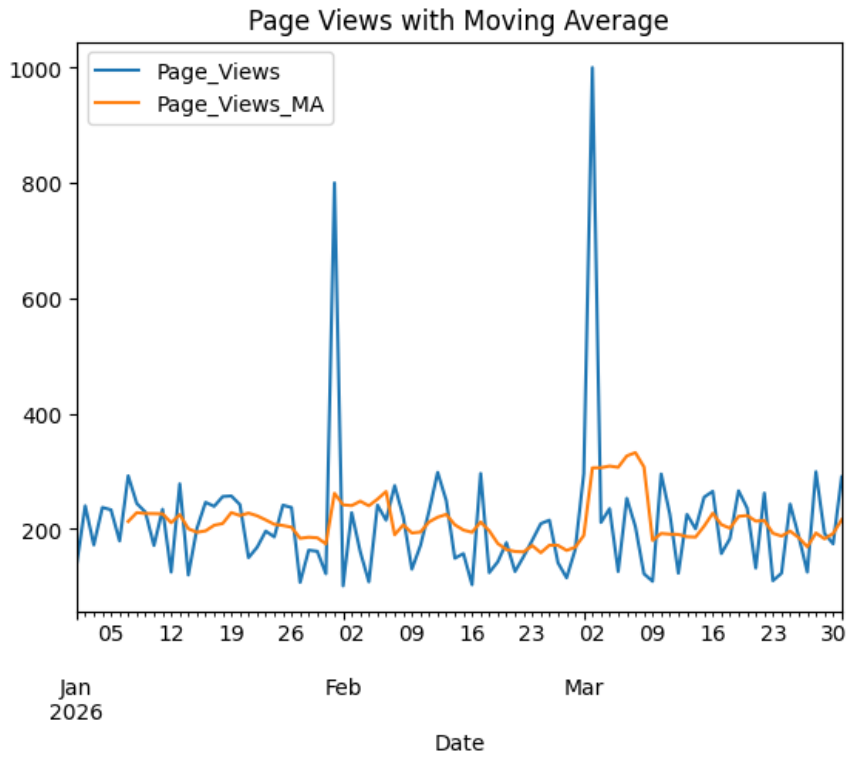
Program:

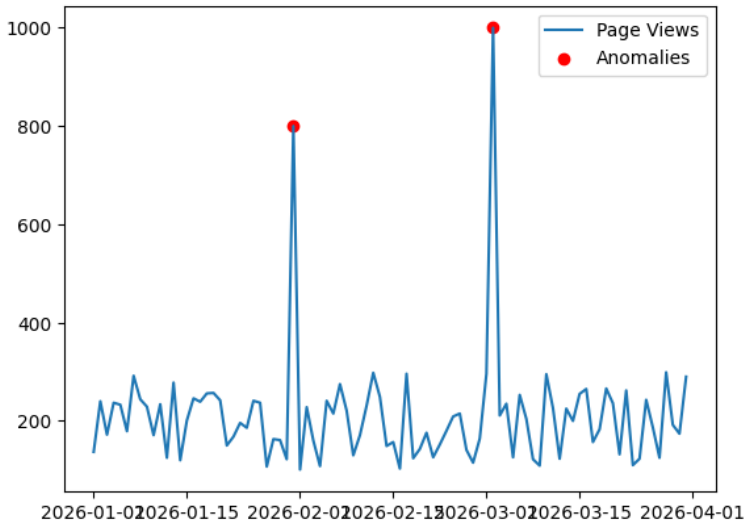
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
np.random.seed(1)
dates = pd.date_range('2026-01-01', periods=90, freq='D')
data = pd.DataFrame({
    'Date': dates,
    'Page_Views': np.random.randint(100, 300, 90),
    'Unique_Visitors': np.random.randint(50, 200, 90),
    'Bounce_Rate': np.random.randint(20, 80, 90)
})
data.set_index('Date', inplace=True)
data.iloc[30, 0] = 800
data.iloc[60, 0] = 1000
decomposition = seasonal_decompose(
    data['Page_Views'], model='additive', period=7
)
decomposition.plot()
plt.show()
data['Page_Views_MA'] = data['Page_Views'].rolling(7).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()
```

```
data['Page_Views_Z'] = (
    data['Page_Views'] - data['Page_Views'].mean()
) / data['Page_Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'],
            color='red', label='Anomalies')
plt.legend()
plt.show()
```

Output:







Result:

Thus, Time series analysis of website traffic was performed successfully.